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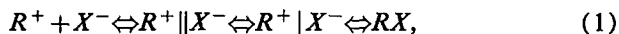
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Classical molecular dynamics computer simulations have been used to investigate the thermodynamics and kinetics of sodium chloride association in polarizable water. The simulations make use of the three-site polarizable water model of Dang [J. Chem. Phys. **97**, 2659 (1992)], which accurately reproduces many bulk water properties. The model's static dielectric constant and relaxation behavior have been calculated and found to be in reasonable agreement with experimental results. The ion-water interaction potentials have been constructed through fitting to both experimental gas-phase binding enthalpies for small ion-water clusters and to the measured structures and solvation enthalpies of ionic solutions. Structural properties and the potential of mean force for sodium chloride in water have been calculated. In addition, Grote-Hynes theory has been used to predict dynamical features of contact ion-pair dissociation. All of the calculated ionic solution properties have been compared with results from simulations using the extended simple point charge (SPC/E), nonpolarizable water model [J. Phys. Chem. **91**, 6296 (1987)]. The dependence on polarizability is found to be small, yet measurable, with the largest effects seen in the solvation structure around the highly polarizable chlorine anion. This work validates the use of some nonpolarizable water models in simulations of many condensed-phase systems of chemical and biochemical interest.

I. INTRODUCTION

The kinetics and thermodynamics of ionic association in solution have been subjects of considerable experimental, theoretical, and computational interest in recent years. Much of this interest results from the important role that ionic association plays in chemical reactions in solution.^{1,2} A general scheme for the reaction of two ionic solutes may be written as



in which RX is the product of a reaction between the ions R^+ and X^- . The intermediate states $R^+ | X^-$ and $R^+ || X^-$ represent a contact ion pair (CIP) and a solvent-separated ion pair (SSIP), respectively. Winstead and co-workers³ were the first to suggest that CIP and SSIP states might act as reaction intermediates, although their existence was independently suggested by Sadek and Fuoss.⁴ Despite this long history of study, there is still considerable uncertainty regarding the specific role these ion-pair states play in chemical reactions in solution.

Increasingly accurate experimental measurements of the relative populations of CIP and SSIP states have been made in recent years using infrared and Raman spectroscopy on a variety of nitrate⁵⁻⁷ and thiocyanate solutions.⁸ It is interesting that for some systems the CIP stability is at a minimum near room temperature, becoming more stable as the temperature is both increased⁶ and decreased.⁷ Experimental measurements of the dynamics of ion-pair interconversion are limited, although recent ultrafast spectroscopy experiments are beginning to provide some useful information.⁹⁻¹²

In the present paper, we use computer simulations to focus on the thermodynamics and kinetics of interconversion of the CIP and SSIP states. In recent years there have

been many integral equation¹³⁻¹⁵ and computational¹⁶⁻²⁹ studies of ionic association. The present simulations differ from previous treatments in that solvent and solute polarizabilities are included explicitly in the simulation model (see, however, Ref. 29). This makes the current study valuable for two reasons. First, the polarizable water model used here³⁰ (labeled RPOL water) is effectively transferable from the gas phase to bulk solution.³¹ Experimental gas-phase data may therefore be incorporated into the development of reliable ion-water interaction models. This is useful given the known sensitivity of ionic association free energies to both ion-water and water-water interactions.^{16,24} The second advantage of explicitly including polarizability is that it allows for a direct evaluation of the importance of polarization fluctuations in 1-1 electrolyte solutions. All of the calculations described in this paper for polarizable water have been repeated using the extended simple point charge (SPC/E) nonpolarizable water model of Berendsen and co-workers.³² Through a comparison of these two model systems, it is possible to evaluate critically the need to explicitly treat molecular polarizability in systems of this type. This work therefore has implications for a wide variety of simulations, many of biological interest, in which nonpolarizable water models are commonly used.

The outline of the paper is as follows. Details of the simulation models and methods are summarized in Sec. II. Properties of the polarizable and nonpolarizable water models, including dielectric properties, are presented and compared in Sec. III. Structural and thermodynamic properties of the two solutions, including ion association free energies, are presented in Sec. IV, and dynamical properties are discussed in Sec. V. The conclusions are summarized and discussed in Sec. VI.

TABLE I. Simulation parameters.

Model	Atom type	σ (Å)	ϵ (kcal mol ⁻¹)	q (e)	α (Å ³)
SPC/E ^a	O	3.166	0.1554	-0.8476	...
	H	...	...	0.4238	...
RPOL	O	3.196	0.1600	-0.7300	0.528
	H	...	...	0.3650	0.170
	Na ⁺	2.350	0.1300	1.0000	0.240
	Cl ⁻	4.450 ^b	0.1000	-1.0000	3.250

^aReference 32.^b4.400 in SPC/E simulations.

II. SIMULATION DETAILS

A. Polarizable interaction model

The RPOL water model developed by Dang³⁰ is a revised version of the model due to Caldwell, Dang, and Kollman.³³ It has three interaction sites and the same rigid tetrahedral geometry as the simple point charge (SPC) models.^{32,34} There are fixed charges and point polarizabilities attached to each atomic site, along with a Lennard-Jones (LJ) interaction centered on the oxygen. Each ion in the aqueous solution simulations is represented similarly by a fixed charge plus point polarizability and a LJ term. The ion and water parameters used in the present study are summarized in Table I. Ion-water and ion-ion cross terms are defined using the usual sum rules. The water parameters are taken directly from Dang,³⁰ while the procedure for choosing the ion parameters is described below.

The total potential energy for the aqueous system is

$$U_{\text{tot}} = U_{\text{pair}} + U_{\text{pol}}, \quad (2)$$

where the pair-additive part of the potential is a sum of LJ and Coulomb interactions

$$U_{\text{pair}} = \sum_i \sum_j \left\{ 4\epsilon \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right] + \frac{q_i q_j}{r_{ij}} \right\}. \quad (3)$$

The polarization energy is given by

$$U_{\text{pol}} = -\frac{1}{2} \sum_i \mu_i \cdot \mathbf{E}_i^0, \quad (4)$$

where μ_i is the induced dipole moment on atom site i and $\mathbf{E}_i^0$ is the electric field at site i due to the fixed charges in the system. The dipole moments are determined at each step in the simulation by iteratively solving the equations

$$\mu_i = \alpha_i \mathbf{E}_i \quad (5)$$

and

$$\mathbf{E}_i = \mathbf{E}_i^0 + \sum_{j \neq i} \mathbf{T}_{ij} \mu_j, \quad (6)$$

where α_i is the atomic polarizability and $\mathbf{T}_{ij}$ is the dipole tensor, defined elsewhere.³³ A third-order predictor algorithm is used to generate an initial estimate of μ_j , and convergence of the dipole moments to within a tolerance of 0.01 D is usually achieved within two iterations.

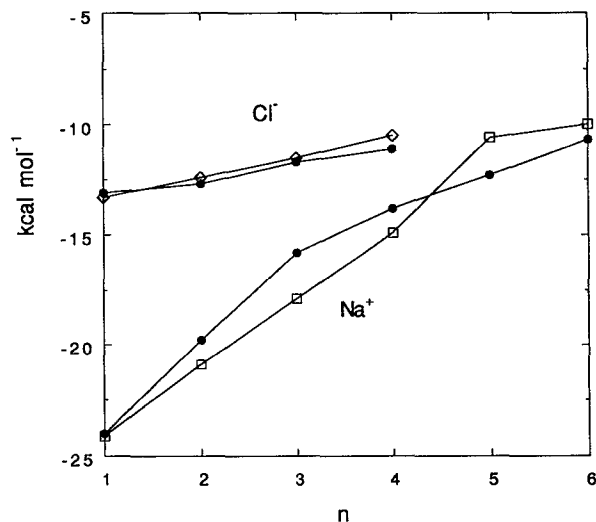


FIG. 1. Calculated incremental binding enthalpies in kcal mol⁻¹ for the gas-phase reactions $\text{Na}^+(\text{H}_2\text{O})_{n-1} + \text{H}_2\text{O} \rightleftharpoons \text{Na}^+(\text{H}_2\text{O})_n$ (squares), and $\text{Cl}^-(\text{H}_2\text{O})_{n-1} + \text{H}_2\text{O} \rightleftharpoons \text{Cl}^-(\text{H}_2\text{O})_n$ (diamonds). The experimental values from Refs. 36 and 37 are also shown as solid circles.

B. Ion-water interaction parameters

In defining the sodium and chloride interaction parameters the ion polarizabilities are first fixed to be approximately equal to experimentally derived values.³⁵ The ion-water LJ parameters are then determined by fitting to gas-phase binding enthalpy data for small sodium³⁶ and chloride³⁷ ion-water clusters. Incremental cluster binding enthalpies for the gas-phase reaction



may be determined from simulations by measuring the average cluster potential energies as a function of cluster size. The enthalpy and energy change for the reaction are related by

$$\Delta H = \Delta U + p\Delta V = \Delta U - RT. \quad (8)$$

Incremental binding enthalpies for the clusters $\text{Na}^+(\text{H}_2\text{O})_n$ ($n=1,6$) and $\text{Cl}^-(\text{H}_2\text{O})_n$ ($n=1,4$) are determined here from a series of 1 ns trajectories. The simulations are performed at constant total energy such that the average kinetic temperatures are near 300 K. The cluster center-of-mass translational and rotational motions are removed at the start of each trajectory.

The incremental binding enthalpies for the optimized ion-water interaction parameters are plotted in Fig. 1. The agreement between experimental and calculated enthalpies is very good for the chlorine ion but more limited for sodium. A modest improvement for sodium can be achieved with the inclusion of an explicit three-body repulsive interaction.³⁸ Nevertheless, we have not included these interactions here because the additional complexity would make the comparison with nonpolarizable water models more difficult. The discrepancies between the experimental and calculated binding enthalpies are small enough that we believe this model should give an effective representation of

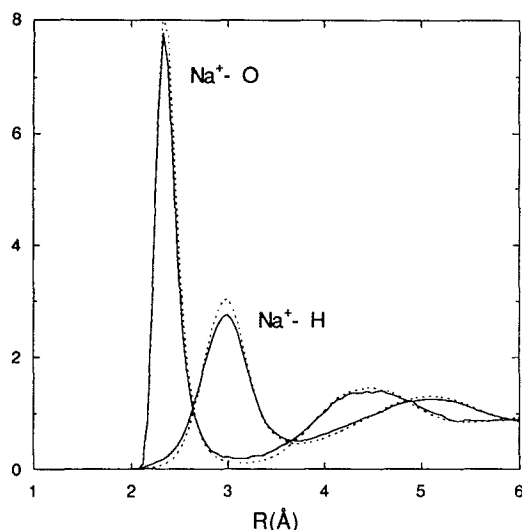


FIG. 2. $\text{Na}^+\text{-O}$ and $\text{Na}^+\text{-H}$ radial distribution functions for RPOL water (solid lines) and SPC/E water (dotted lines).

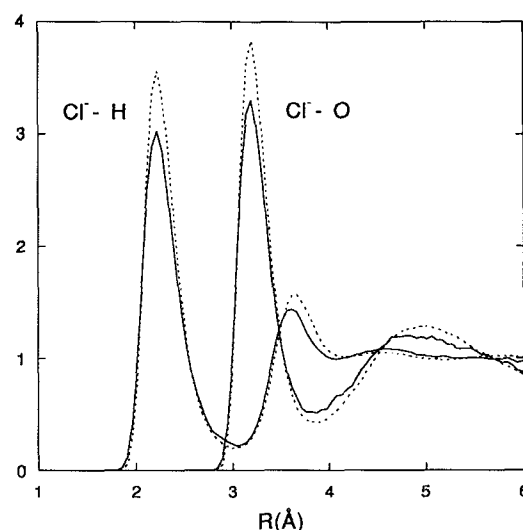


FIG. 3. $\text{Cl}^-\text{-O}$ and $\text{Cl}^-\text{-H}$ radial distribution functions for RPOL water (solid lines) and SPC/E water (dotted lines).

sodium chloride solutions. This claim is strengthened below through comparison with experimental structures and hydration enthalpies for aqueous sodium chloride solutions.

Liquid-phase structures and hydration enthalpies have been determined for the optimized sodium and chloride parameters from a simulation of a sodium chloride ion pair in 214 RPOL water molecules. For this calculation the ion pair separation has been fixed at various distances greater than or equal to 6.0 Å (the same trajectories have been used in the potential of mean force calculation described later). The ion-water structures should therefore resemble the structures around isolated ions. The volume of the cubic simulation cell is held fixed with a side length of 18.63 Å. Long range charge-charge, charge-dipole, and dipole-dipole interactions are calculated using the Ewald summation method.³⁹ A simulation time step of 1.0 fs is employed, and the total simulation time is 160 ps.

The calculated ion-water radial distribution functions for Na^+ and Cl^- are displayed in Figs. 2 and 3, respectively. The simulated peak positions and coordination numbers are summarized in Table II along with the available experimental data. Agreement with experimental results is excellent. The hydration energy for the sodium chloride pair is calculated by measuring the total potential energy of the system and subtracting the energy of 214 bulk water molecules. The resulting solvation energy is $-182 \pm 3 \text{ kcal mol}^{-1}$, which is in reasonable agreement with the experimental solvation enthalpy of $-188 \text{ kcal mol}^{-1}$.⁴⁰ Thus, we see that the present model gives a good representation of the experimentally derived properties of aqueous sodium chloride solutions.

Finally, the ion-SPC/E water interaction parameters are defined such that the ion-water solvation structures in the liquid phase closely match the RPOL model predictions. In particular, the interaction parameters are defined such that the ion-hydrogen and ion-oxygen first solvation

shell peak positions are the same for the two models. This requires no modification of the sodium ion parameters, and a slight decrease in the chlorine ion size parameter. The resulting ion-SPC/E water structures are also displayed in Figs. 2 and 3. Structural peaks are heightened and narrowed somewhat in SPC/E water, and there is a net increase in the calculated coordination numbers. This effect is larger for Cl^- , where the ion's polarizability is known to lead to a reduction in its average coordination number.⁴¹

III. BULK WATER PROPERTIES

The RPOL water model was derived originally from the SPC water models,^{32,34} and maintains their characteristic tetrahedral geometry. We have chosen to compare the properties of the RPOL model and its solutions with the best available rigid, nonpolarizable water model of comparable geometry, the SPC/E model of Berendsen and co-workers.³² This model improves upon the original SPC model³⁴ through the addition of a self-polarization energy correction.

The bulk properties of the RPOL and SPC/E models have been calculated from simulations of 216 water molecules in a cubic simulation box at a constant density of

TABLE II. Ion-water structural properties.

	RPOL	SPC/E	Experiment
$R(\text{Na}^+\text{-O})$, Å	2.33	2.33	2.38, 2.4 ^a
$R(\text{Na}^+\text{-H})$, Å	3.00	2.98	...
$\text{Na}^+\text{-O}$ coordination No.	5.6	5.8	4, 6 ^a
$R(\text{Cl}^-\text{-O})$, Å	3.20	3.20	3.1 ± 0.1^b
$R(\text{Cl}^-\text{-H})$, Å	2.22	2.22	2.28 ± 0.03^b
$\text{Cl}^-\text{-H}$ coordination No.	6.5	6.9	6.4 ± 0.3^b

^aG. W. Neilson and J. E. Enderby, *Annu. Rep. Prog. Chem., Sect. C* **76**, 185 (1979).

^bD. H. Powell, G. W. Neilson, and J. E. Enderby, *J. Phys.: Condens. Matter* **5**, 5723 (1993).

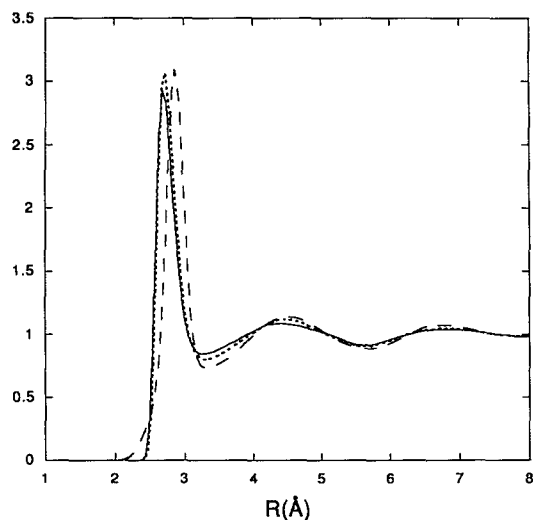


FIG. 4. The O-O radial distribution function for RPOL water (solid line) and SPC/E water (dotted line), along with experimental results (dashed line) are taken from Ref. 43.

0.997 g cm^{-3} . Long range charge-charge and, for the polarizable system, charge-dipole, and dipole-dipole interactions are calculated using the Ewald summation technique.³⁹ A constant temperature of 298 K is maintained through "weak coupling"⁴² to an external bath with a time constant of 0.4 ps. Time steps of 1.0 and 1.5 fs and total simulation times of 500 and 600 ps are used for the RPOL and SPC/E simulations, respectively.

The oxygen-oxygen radial distribution functions for the two models are displayed in Fig. 4, along with the experimental data of Soper and Phillips.⁴³ Agreement between the SPC/E and RPOL models is good, although the peak height is somewhat greater for SPC/E water. Both models show substantial deviation from the experimental O-O peak position. Oxygen-hydrogen and hydrogen-hydrogen radial distribution functions, not shown, are both in good agreement with experimental results.

The bulk properties of the RPOL and SPC/E water models are summarized in Table III. The mean interaction energy, pressure, and diffusion constant of the two models

TABLE III. Bulk water properties at 298 K and 0.997 g cm^{-3} .

	RPOL	SPC/E	Experiment
U_{pot} (kcal mol ⁻¹)	-9.90	-9.90	-9.92
P (kbar)	-0.07 ± 0.04	-0.08 ± 0.04	0.001
D ($\times 10^{-9} \text{ m}^2 \text{ s}^{-1}$)	2.4 ± 0.3	2.4 ± 0.4	2.3^a
ϵ	106 ± 18	67 ± 10	78.3^b
τ_D (ps)	11 ± 4	10 ± 3	8.1^b
τ_1^+ (ps)	5.5 ± 0.2	5.1 ± 0.2	...
τ_2^+ (ps)	2.7 ± 0.1	2.8 ± 0.1	...
τ_{NMR} (ps)	1.8 ± 0.1	1.9 ± 0.1	2.5^c

^aK. Krynicki, C. D. Green, and D. W. Sawyer, *Faraday Discuss. Chem. Soc.* **66**, 199 (1978).

^bD. Bertolini, M. Cassettari, and G. Salvetti, *J. Chem. Phys.* **76**, 3285 (1982).

^cJ. Jonas, T. DeFries, and D. J. Wilbur, *J. Chem. Phys.* **65**, 582 (1976).

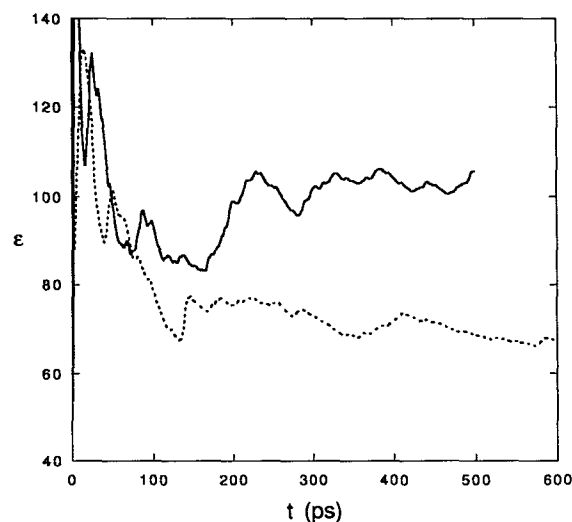


FIG. 5. Running averages of the static dielectric constant for RPOL water (solid line) and SPC/E water (dotted line).

are in almost exact agreement with experimental values. Tabulated uncertainties are calculated here and throughout by a statistical analysis of block averaged data, with block sizes of 20 and 30 ps for RPOL and SPC/E water, respectively.

The static dielectric constant is related to the fluctuations in the total box dipole moment, $\mathbf{M}$, by⁴⁴

$$\epsilon = \frac{4\pi\beta}{3V} [\langle \mathbf{M}^2 \rangle - \langle \mathbf{M} \rangle^2] + \epsilon_\infty, \quad (9)$$

where $\beta = (k_b T)^{-1}$. The high frequency dielectric constant, ϵ_∞ , should be small compared to the magnitude and uncertainty in the dipole fluctuation term in Eq. (9),⁴⁵ and we approximate it by a nonpolarizable model value of unity. In addition, the average dipole moment for the simulation box, $\langle \mathbf{M} \rangle$, should be zero for an isotropic system. We therefore calculate the dielectric constant from

$$\epsilon - 1 = \frac{4\pi\beta}{3V} \langle \mathbf{M}^2 \rangle, \quad (10)$$

while the ratio of $\langle \mathbf{M} \rangle^2$ to $\langle \mathbf{M}^2 \rangle$ acts as a measure of convergence. Running averages of the static dielectric constant calculated in this way are displayed in Fig. 5. The dielectric constant for SPC/E water is somewhat smaller than the experimental value of 78.3, and is in good agreement with a previous calculation.⁴⁶ The dielectric constant for RPOL water is, in contrast, too large, though still in reasonable agreement with the experimental value. The convergence of each calculation can be estimated from both the tabulated uncertainties and from the ratios of $\langle \mathbf{M} \rangle^2$ to $\langle \mathbf{M}^2 \rangle$, which are 0.03 and 0.05 for RPOL and SPC/E water, respectively.

Although the static dielectric constants of the two water models are substantially different, their dielectric relaxation times are similar and in good agreement with the

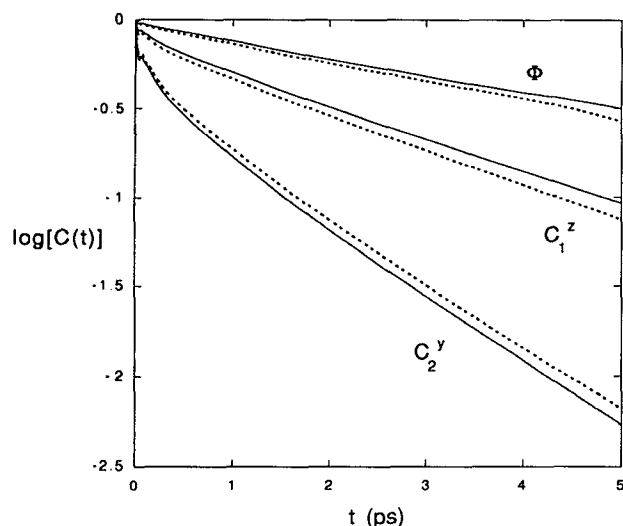


FIG. 6. The collective dipole moment correlation function $\Phi(t)$ and two individual molecule orientational correlations functions, $C_1^z(t)$ and $C_2^y(t)$ for RPOL water (solid lines) and SPC/E water (dotted lines).

experimental value. The value of τ_D is calculated from the decay time of the collective dipole moment correlation function

$$\Phi(t) = \frac{\langle \mathbf{M}(t) \cdot \mathbf{M}(0) \rangle}{\langle \mathbf{M}^2 \rangle}. \quad (11)$$

The log of $\Phi(t)$ for both water models is displayed in Fig. 6. Two individual molecule orientational correlation functions are also shown in the figure, namely,

$$C_1^z = \langle P_1[\mathbf{e}^z(t) \cdot \mathbf{e}^z(0)] \rangle \quad (12)$$

and

$$C_2^y = \langle P_2[\mathbf{e}^y(t) \cdot \mathbf{e}^y(0)] \rangle, \quad (13)$$

where P_l is the l th Legendre polynomial and $\mathbf{e}^\alpha$ is the unit vector in the α direction in the coordinate frame of the molecule. The z axis is the dipole moment axis, and the y axis is parallel to a vector connecting the two hydrogen atoms. The long-time exponential decay times for these two correlation functions give the values of τ_1^z and τ_2^y shown in Table II. Once again, the two models are in excellent agreement. Both the SPC/E and RPOL models slightly underestimate the experimental NMR relaxation time, which is related directly to the integrated area of C_2^y .⁴⁷

In summary, the RPOL and SPC/E water models are similar with respect to all structural, dynamical, and thermodynamic properties with the largest differences observed for the static dielectric constant. A comparison of properties of solutions of these two models is therefore a valid method for evaluating the role polarization fluctuations play in various aqueous solutions.

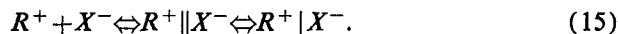
IV. IONIC ASSOCIATION THERMODYNAMICS

A. The potential of mean force

The solvent contribution to the free energy of reaction of two ions may be decomposed into two contributions. The first is the relative solvation free energy of reactants and products in the reaction



where, as above, $R^+ | X^-$ corresponds to the CIP reactants and RX to the products. The second contribution, which we evaluate here, is the association free energy required to bring the ionic reactants from large separation into contact,



Here the species $R^+ || X^-$ corresponds to an intermediate SSIP configuration. The free energy of the ion pair as a function of ionic separation is known as the potential of mean force, or PMF. Its shape reveals many interesting details about the solvation of ionic species in solution, and is relevant to both the overall thermodynamics and the kinetics of ionic reactions.

We have calculated the PMF for a sodium chloride ion pair in both RPOL and SPC/E water using a standard thermodynamic integration technique.⁴⁸ At each step in the calculations, the ion-pair separation is fixed and the average force along the ion-pair axis is calculated as a function of distance. This average force is subsequently integrated to give the PMF. For these calculations, the ion-pair separations have been fixed at distances between 2.50 and 7.75 Å, with a separation increment of 0.25 Å. The total simulation time at each ion-pair separation is at least 150 ps for SPC/E water and 40 ps for RPOL water. Substantially longer averaging is performed for RPOL water in the region of the CIP and its dissociation barrier, as well as in the 5.50–6.00 Å region. Table IV contains a summary of the simulation times and average forces for all separations. Uncertainties are also reported for selected separations where the block averaging technique has been applied.

The sodium chloride PMF for RPOL water is displayed in Fig. 7. The absolute free energy of the PMF is not determined directly from the simulation, but is defined instead by matching to the primitive model prediction at large separation. In the primitive model, the solvent is assumed to act as a dielectric continuum and the ion-pair interaction at infinite dilution is given by the gas-phase potential with the Coulomb interactions scaled by the static dielectric constant. The primitive model result shown in the figure is determined using the static dielectric constant of RPOL water calculated above to be 106. Given this absolute free energy assignment, the CIP configuration exhibits a free energy stabilization of 1.3 kcal mol⁻¹ relative to the infinitely separated ions. The barrier to CIP dissociation is 1.8 kcal mol⁻¹, and the SSIP minimum is 0.2 kcal mol⁻¹ less stable than the CIP minimum. Previous calculations of the sodium chloride PMF have varied substantially in their predictions of the relative stability of CIP and SSIP configurations,^{16,24–26} at least in part due to vari-

TABLE IV. Average forces.

Distance	RPOL		SPC/E	
	Time ps	Ave. force kcal mol ⁻¹ Å ⁻¹	Time ps	Ave. Force kcal mol ⁻¹ Å ⁻¹
2.50	40	44.0	150	49.3
2.75	100	2.4	150	4.0
3.00	100	-3.7	150	-4.2
3.25	157	-3.2	150	-3.5
3.50	340	-1.4±0.2	450	-1.2±0.2
3.75	340	0.5±0.3	450	1.2±0.4
4.00	40	1.3	150	1.9
4.25	40	2.1	150	2.8
4.50	40	1.7	150	1.2
4.75	40	0.7	150	0.6
5.00	40	0.5	150	-0.3
5.25	40	-0.5	150	-0.2
5.50	100	-0.7	150	-1.4
5.75	220	-0.8±0.3	450	-0.8±0.2
6.00	130	0.1	150	0.0
6.25	40	-0.6	150	0.3
6.50	40	0.0	150	0.3
6.75	40	-0.4	150	-0.1
7.00	40	-0.2	150	-0.1
7.25	40	0.0	150	-0.4
7.50	40	0.2	150	-0.1
7.75	40	-0.6	150	-0.1

ations in both the ion-water and water-water interactions employed. We believe that the current results are of improved reliability because of the connection of the interaction parameters with gas-phase binding enthalpy data as described above.

The PMF for sodium chloride in SPC/E water is also displayed in Fig. 7, illustrating the effect of polarization fluctuations on the PMF. The difference in primitive model results for RPOL and SPC/E water is a direct consequence

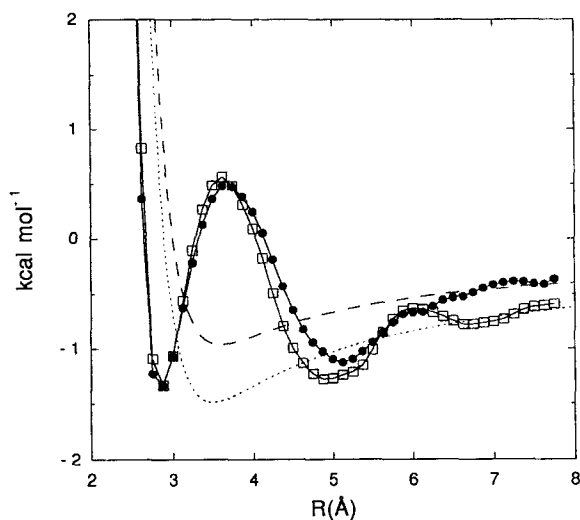


FIG. 7. The NaCl potential of mean force in RPOL water (filled circles) and SPC/E water (open squares). The primitive model predictions are also shown for continuum solvents with static dielectric constants corresponding to RPOL (dashed line) and SPC/E (dotted line) water.

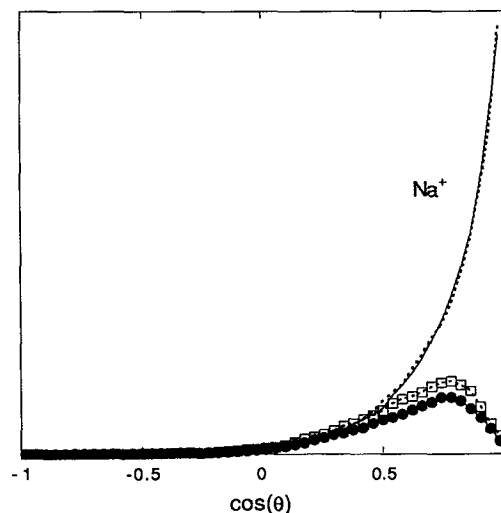


FIG. 8. First solvation shell orientational correlations functions around Na⁺. Bridging water molecules are represented with filled circles (RPOL water) and open squares (SPC/E water). Other coordinated water molecules are represented by solid lines (RPOL water) and dotted lines (SPC/E water).

of the difference in calculated dielectric constants for the two models. Other than this effect at large separations, the two PMFs are almost identical. For RPOL water, the position of the CIP dissociation barrier appears to be shifted out slightly, the SSIP minimum is shifted out and up slightly, and the SSIP configuration is less well defined. Although these observed differences may be real, the substantial uncertainty in the free energy calculations makes definitive conclusions difficult. In summary, it appears that solvent and solute polarization fluctuations do not have any dramatic effect on the thermodynamics of ionic association in this system.

B. Structure

Several structural and dynamical aspects of the ionic association process have also been investigated. These studies have focused primarily on the barrier region between the CIP and SSIP configurations. In particular, lengthy calculations (>340 ps) have been performed in both solvents with the ion-pair separation fixed at 3.75 Å. This configuration is very near to the peak in the PMFs shown in Fig. 7. The solvation structure around this configuration is described in this section, while dynamical aspects of both hydration and ionic association will be discussed in Sec. V.

Oriental correlation functions for water molecules in the first solvation shell of the sodium and chloride ions are displayed in Figs. 8 and 9, respectively. A water molecule is defined as being coordinated to the sodium ion if $r_{\text{Na-O}} < 3.20$ Å, and to the chloride ion if $r_{\text{Cl-H}} < 3.05$ Å. Bridging water molecules are coordinated to both ions simultaneously. The orientation of the water molecules is defined in terms of the angle θ between the ion-oxygen vector and the water HOH bisector. The correlation functions are normalized such that their integrated areas rep-

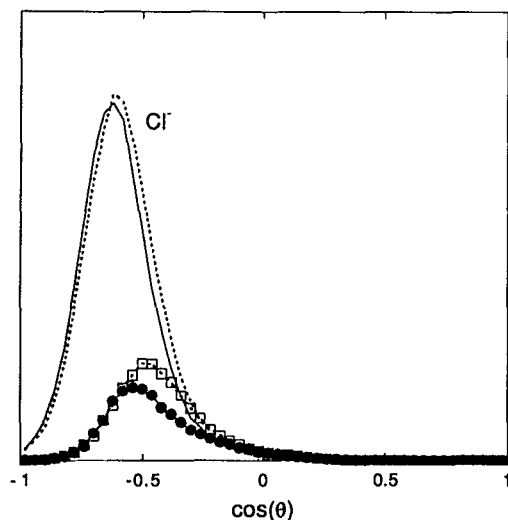


FIG. 9. As in Fig. 8 except that the correlation functions are around Cl^- .

resent relative populations. It is evident from the figures that bridging water molecules have substantially perturbed orientations compared with singly coordinated molecules. This is a symptom of the strained solvent configurations around the barrier region. As the ion-pair separation is increased the bridging water orientations relax toward their more optimal, singly coordinated distributions. The RPOL and SPC/E orientational correlation functions differ primarily in that there is a net reduction in the hydration number around the ion pair in polarizable water. This is consistent with the hydration structure observed around isolated ions as described in Sec. II above. The average orientation of the bridging water molecules also varies somewhat as a function of polarizability, with this being most evident in Fig. 9.

There is a continual fluctuation in the number of bridging water molecules, N_b , observed during the course of these simulations. The correlation between N_b and the force along the ion-pair axis for $r_{\text{NaCl}} = 3.75 \text{ \AA}$ is displayed in Fig. 10. The behavior illustrated in the figure is not surprising; as water molecules insert themselves between the ions, there is an average force driving the ions apart. The average number of bridging water molecules is reduced from 1.50 in SPC/E water to 1.15 in RPOL water. This difference may account in part for the measured differences in mean force at $3.75 \text{ \AA}$ in the two different solvents (see Table IV).

V. DYNAMICAL PROPERTIES

A. Hydration shell residence times

The aqueous coordination of monovalent ions involves many interesting dynamical properties in addition to the average coordination structures discussed above. The common definition of a coordination shell—the number of water molecules within a certain radius about an ion—is misleading in the sense that it ignores these dynamics. For example, coordinated water molecules may make short ex-

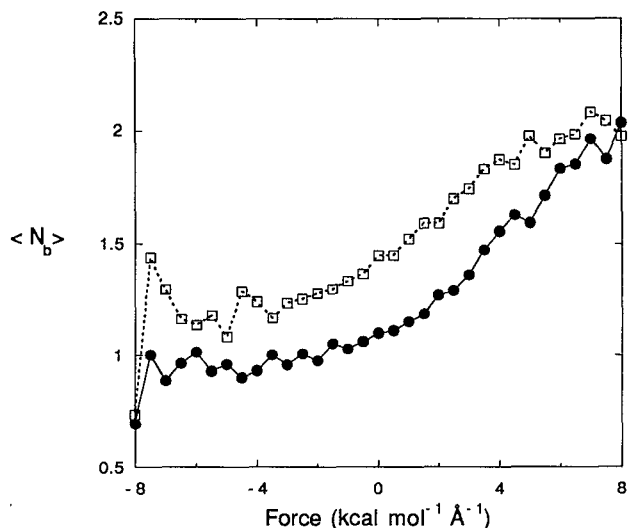


FIG. 10. Correlation of the number of bridging water molecules with the force along the ion-pair axis for RPOL water (filled circles) and SPC/E water (open squares).

cursion out of the coordination region without actually forming any new coordination with the solvent. Likewise, “bulk” water molecules may enter the coordination region without forming any strong ion–water interactions. More complete definitions of coordination therefore may be formulated by taking into account either the time scale or the energetics of hydration. To illustrate some of the dynamical features of ionic hydration we have calculated a hydration correlation function defined by⁴⁹

$$n(t-t_0) = \left\langle \sum_j P_j(t, t_0; t^*) \right\rangle, \quad (16)$$

where $P_j(t, t_0; t^*) = 1$ if water molecule j is in an ion's first coordination shell at both time t_0 and t , and if the molecule has left that coordination shell for no continuous period longer than t^* in the interim. Otherwise, $P_j(t, t_0; t^*) = 0$. The usual coordination number is equal to $n(0)$. The value of t^* is taken to be 2.0 ps, the approximate residence time of water coordinated with water, in order to be consistent with previous calculations.⁴⁹

The hydration correlation functions around the sodium and chloride ions at $3.75 \text{ \AA}$ separation are shown in Fig. 11. The correlation functions are nearly exponential except for times ≤ 0.5 ps, with this short-time feature resulting from brief incursions of bulk water molecules into the coordination region. The hydration shell residence time, τ_R , is defined as the exponential decay time of the hydration correlation function. In RPOL water, $\tau_R = 29$ ps around the sodium ion and 12 ps around the larger chloride ion. These residence times are substantially longer than those calculated previously for a different water model,⁴⁹ but are in reasonable agreement with the poorly characterized experimental values.⁵⁰ The measured residence times are sensitive to the assumed value of t^* . For example, for $t^* = 0$ the residence times are 15 and 3.1 ps around the sodium and chloride ions, respectively. This

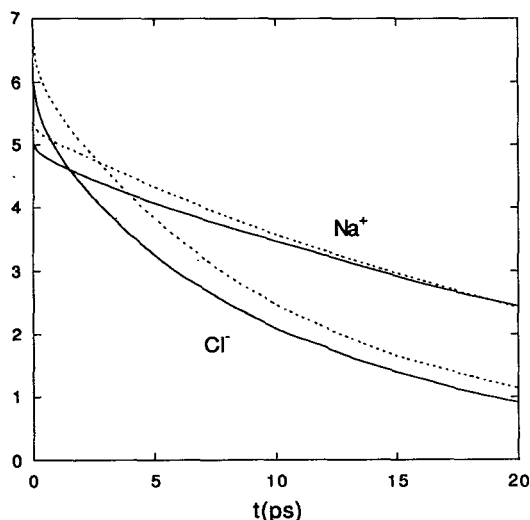


FIG. 11. Hydration correlation functions for the first solvation shell water molecules around Na^+ and Cl^- . Solid lines are for RPOL water and dotted lines for SPC/E water.

sensitivity illustrates the common tendency of coordinated water molecules to travel away from the coordination region for short periods of time. Finally, the measured residence times in SPC/E water are typically about 10% shorter than in polarizable water. The dependence on solvent and solute polarizability is therefore once again seen to be small.

B. CIP dissociation dynamics: Grote-Hynes theory

The dynamics of interconversion of CIP and SSIP configurations is of central importance to the kinetics of ion-pair reactions in solution. The calculation of dynamical rates may be performed conveniently through the use of a constrained molecular dynamics technique,⁵¹ in which trajectories are initiated near the barrier maximum and the motion is traced both forward and backward in time. A calculation of this type has been performed previously by Rey and co-workers²⁸ for NaCl in a flexible version of SPC water. Their results give nearly exact agreement with the predictions of the Grote-Hynes (GH) theory,⁵² a rate theory based on the generalized Langevin equation. Given the good agreement between GH theory predictions and explicit rate calculations, we have chosen to investigate the dynamics of CIP to SSIP interconversion in the context of the GH theory.

Classical transition state theory (TST) predicts that the rate constant for CIP dissociation is a function only of the temperature, the ion-pair reduced mass, and the PMF in the CIP region. The TST forward rate constant for this reaction is given explicitly by⁵³

$$k_f^{\text{TST}} = (2\pi\beta\mu_i)^{-1/2} \frac{(r^\ddagger)^2 e^{-\beta W(r^\ddagger)}}{\int_0^{r^\ddagger} dr r^2 e^{-\beta W(r)}}, \quad (17)$$

where $r^\ddagger$ is the position of the barrier maximum, $\beta = (k_B T)^{-1}$, and μ_i is the ion pair reduced mass. The

TABLE V. Rate theory results.

	RPOL	SPC/E
$k_{\text{TST}} \text{ (ps}^{-1}\text{)}$	0.30	0.29
$\omega_b \text{ (ps}^{-1}\text{)}$	15.5	16.0
κ_{GH}	0.29	0.17
$k = \kappa_{\text{GH}} k_{\text{TST}} \text{ (ps}^{-1}\text{)}$	0.087	0.048

calculated TST rate constants for NaCl dissociation in RPOL and SPC/E water are nearly identical and are presented in Table V. It is well known that the predictive ability of TST breaks down in solution due to dynamical coupling to the solvent in the reaction barrier region. The GH theory is an attempt to correct for this solvent coupling by using a generalized Langevin equation model for the solvent. The forward rate constant for CIP dissociation predicted by GH theory is

$$k_f = \kappa_{\text{GH}} k_f^{\text{TST}}, \quad (18)$$

where κ_{GH} is a transmission coefficient that takes on values between zero and unity. The GH transmission coefficient is given by

$$\kappa_{\text{GH}} = \left(\kappa_{\text{GH}} + \int_0^\infty dt \frac{\zeta(t)}{\mu_i \omega_b} e^{-\omega_b \kappa_{\text{GH}} t} \right)^{-1}, \quad (19)$$

where ω_b is the imaginary barrier frequency at the transition state and $\zeta(t)$ is the time-dependent friction exerted on the reaction coordinate. The transmission coefficient depends on the Fourier-Laplace transform of the time-dependent friction at the frequency $\omega_b \kappa_{\text{GH}}$, and may be determined iteratively if the barrier frequency and friction are known. Values of ω_b calculated from the PMFs are displayed in Table V and are quite similar for the two water models.

For the present case, $\zeta(t)$ is a function of the force exerted along the bond axis when the ion pair separation is fixed at r ,

$$\zeta(t) = \beta [f(t; r) f(0; r)]. \quad (20)$$

The time-dependent friction for both water models with $r = 3.75 \text{ \AA}$ is plotted in Fig. 12. The initial decay of each function is rapid, with a time scale related to the duration of collisions in the system. After some short-time oscillatory behavior, there is a long-time decay that lasts several picoseconds. The behavior beyond 1 ps does not contribute significantly to the rate calculations and is therefore not displayed. Comparison of the results plotted in the figure indicate that the friction is measurably stronger in SPC/E water than in RPOL water, although the two curves are qualitatively similar. The resulting transmission coefficients and forward rate constants are displayed in Table V. Both transmission coefficients are in reasonable agreement with previous calculations for the CIP dissociation of NaCl in water.^{20,27} The variation in the solvent coupling between RPOL and SPC/E water leads to a difference in transmission coefficients and final rate constants of nearly a factor of 2, with the rate being faster in RPOL water. It should be noted, however, that these results are quite sensitive to the

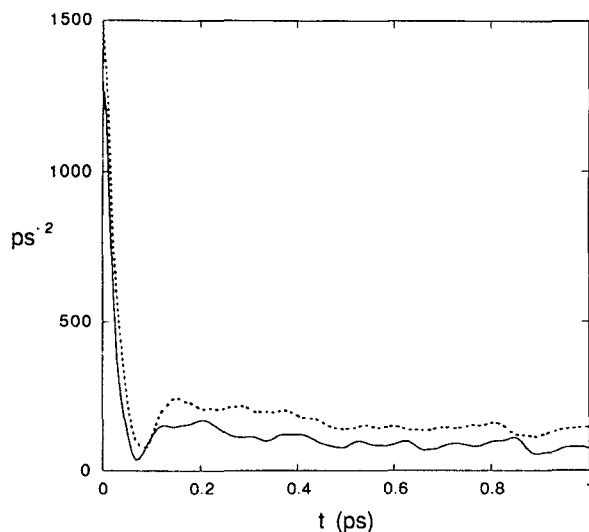


FIG. 12. Time-dependent friction calculated with the ion pair separation fixed at 3.75 Å for RPOL water (solid line) and SPC/E water (dotted line).

details of the PMFs and uncertainties in the final rate constants are comparable to the measured differences.

VI. CONCLUSIONS

Our understanding of the microscopic nature of chemical and biochemical systems has increased substantially in recent years through the application of computer simulation methods. The power of these methods comes from the ability to directly observe every molecular detail of model chemical systems and reactions. This information is meaningful, however, only to the extent that the model interaction potentials employed in the simulations are able to mimic real chemical systems. In this paper, we have presented a thorough study of an aqueous sodium chloride solution using a state-of-the-art polarizable interaction model. The inclusion of polarizability should provide an enhancement in the simulation reliability, compared with pairwise additive model calculations. In addition, we have evaluated critically the ability of one "typical" nonpolarizable model to represent the same system. This work therefore has implications for a wide range of simulations where the use of nonpolarizable interactions models is presently a computational necessity.

The calculated PMF for sodium chloride in RPOL water indicates that contact ion pair and solvent-separated ion pair configurations are of comparable stability and that the barrier to their interconversion is 1.5–2.0 kcal mol⁻¹. Previous calculations of these quantities have varied widely,^{16,24–26} probably because of the use of different interaction models. The reliability of the present calculation is enhanced by both the incorporation of molecular polarizability and by the connection with gas-phase binding enthalpies for ion–water clusters. An investigation of the dynamics of contact ion-pair dissociation using Grote–Hynes theory indicates that solvent coupling reduces the rate con-

stant for dissociation by a factor of 0.29, in reasonable agreement with previously predicted transmission coefficients.^{20,27}

Direct comparison of the properties of NaCl in RPOL and SPC/E water indicates clearly that polarization fluctuations play a minor role in this system. Although the two solutions are measurably different with regard to several structural and dynamical properties, these differences are smaller than the accuracy required or expected from simulations. This result validates the use of nonpolarizable models for simulations of many aqueous solutions. We note, however, that SPC/E water is superior to many pairwise additive models in its static treatment of bulk-water polarization. We recommend its use in particular, and the use of other models that include self-polarization energy corrections. Finally, the development and application of polarizable models for water and aqueous solutions remains a subject of considerable importance. For example, polarizability is likely to be an essential ingredient in achieving transferability from gas phase to condensed phase, as well as in modeling electrode surfaces and divalent-ion hydration.

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